

Super Admin Complete Architecture & Workflow Guide

A comprehensive multi-tenant SaaS operational manual covering platform-level governance, store provisioning, subscription billing, audit security, and system telemetry.

Role: Super Administrator	Target Domain: Platform SaaS Governance	Architecture: Multi-Tenant Laravel + MySQL	Version: 1.0 (Production)
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1. Core Philosophy & Role Separation

Super Admin kisi ek medical store ka manager nahi hai — wo poore multi-tenant SaaS cloud platform ka global owner/administrator hai.

Entity Role	Scope & Operational Boundary	Data Access Isolation
Super Admin	Poore software infrastructure, stores, billing plans, subscriptions, global settings aur compliance audits ka manager.	Cross-tenant global visibility across all medical stores.
Store Owner	Apne specific medical store (pharmacy) ka owner. Inventory, POS billing, purchases, staff aur medicine records chalata hai.	Strict multi-tenant isolation (sirf apne store ka data access kar sakta hai).
Staff (Optional)	Store owner ke under employees (Cashier, Billing operator, Pharmacist). Agar owner akela hai to staff banana compulsory nahi hai.	Owner-delegated permissions inside tenant store.
Customer (Optional)	Walk-in retail customer. Har sale ke liye customer account compulsory nahi hai; credit/prescription tracking ke liye save hota hai.	Store-level customer ledger.

2. Super Admin Modules & Technical Architecture

1. Super Admin Authentication (/super-admin/login)

- **Isolated Entry Point:** Standard store users cannot login via Super Admin portal.
- **No Public Registration:** Super Admin accounts are seeded / protected server-side.
- **Security Rules:** Inactivity session termination, rate limiting, and automated auth audit logging.

2. Executive Super Admin Dashboard (/super-admin/dashboard)

- **Real Database KPIs:** Total Stores, Active Stores, Inactive Stores, Expiring Subscriptions, Monthly Revenue.
- **Recent Directory:** Newly provisioned stores with owner linkage.
- **Live Activity Stream:** Powered by real-time `audit\_logs` model timeline.

3. Medical Store Management (/super-admin/stores)

- **Sequential Code Generator:** Automated unique store code (e.g. MED-000001, MED-000002).
- **Store Directory & Filter:** Search by name/code, city filter, status toggle (Active, Inactive, Suspended).
- **Statutory & Regulatory Records:** GSTIN number, Drug License No, License Expiry Date, Store Type (Retail / Wholesale).
- **Brand Assets:** Store logo uploads stored securely in public disk.

4. Store Owner Management (/super-admin/store-owners)

- **Direct Store Association:** Har owner kisi ek valid store se connected hota hai.
- **Tenant Boundary Guard:** Rahul (Sharma Medical) can never see Gupta Pharmacy's data.
- **One-Click Status Toggle:** Activate / Deactivate owner logins instantly.

5. Subscription Plan Engine (/super-admin/subscriptions/plans)

- **Tiered Packaging:** Basic, Professional, Enterprise, Custom Yearly plans.
- **Dynamic Usage Quotas:** Max Staff limits, Medicine inventory caps, Monthly invoice volume caps, Customer record caps.
- **Feature Toggles:** Inventory, Purchase orders, POS terminal, GST reporting, Advanced analytics.

6. Store Subscription Lifecycle (/super-admin/subscriptions/stores)

- **Plan Assignment:** Medical store ko desired plan, start date, end date aur trial duration assign karna.
- **Automatic Status Transition:** Trial, Active, Expiring, Expired, Cancelled, Suspended.
- **Historical Preservation:** Store ka complete subscription history and renewal timeline securely stored.

7. Revenue & Payment Management (/super-admin/payments)

- **Billing Ledger:** Track subscription payments (Amount, Currency, Date, Status).
- **Multi-Channel Support:** UPI, Online Gateway, Net Banking, Credit/Debit Card, Direct Cash, Bank Transfer.
- **Audited Transaction ID:** Generated transaction reference code (e.g. TXN-XXXXXXXX) with status lifecycle (Paid, Pending, Failed, Refunded).

8. Platform Reports & Analytics (/super-admin/reports)

- **Overview Analytics:** Platform gross revenue, tenant growth velocity, churn rate.
- **Granular Breakdowns:** Store geographic distributions, subscription tier share, payment collections by method.
- **Early Expiration Alerts:** Stores expiring within 7-30 days highlighted for renewal follow-up.
- **Data Portability:** One-click CSV export for external business intelligence and compliance audits.

9. Platform Notification Center (/super-admin/notifications)

- **Target Audience Selector:** Broadcast to All Medical Stores or Target Specific Store.
- **Urgency & Types:** General, Announcement, Subscription Expiry Warning, Payment Update, System Upgrade.
- **Delivery Scheduling:** Instant dispatch ('Send Now') or Scheduled future date/time dispatch.
- **Immutability Guard:** Published/Sent notifications cannot be mutated to preserve communication audit trail.

10. Centralized System Settings (/super-admin/settings)

- **General Settings:** Application Name, Brand Tagline, Default Timezone (Asia/Kolkata), Currency Symbol (₹), Date/Time Formats.
- **Branding Assets:** Platform Header Logo, Tab Favicon, Login Portal Emblem.
- **Corporate Footprint:** SaaS vendor legal name, support email desk, corporate address.
- **System Operations:** Maintenance Mode toggle with customizable tenant downtime message (Super Admin bypass preserved).
- **Security Policies:** Session timeout minutes, password complexity rules, brute-force rate limiter.
- **Instant Cache Sync:** Key-value settings layer with automatic Redis/File cache invalidation upon saving.

11. Immutable Audit Trail System (/super-admin/audit-logs)

- **Activity Ledger:** Chronological security trace of all Super Admin operations.
- **Sensitive Redaction:** Passwords, auth tokens, secrets are permanently stripped before storage.
- **Client Telemetry:** IP address, browser user-agent, target morph subject reference.
- **Field-Level Diff Inspector:** Side-by-side comparison of Previous Value (Old) vs Updated Value (New).

3. Practical End-to-End Onboarding Walkthrough

Maan lo ek naya medical store — 'Sharma Medical Store' (Varanasi) software me join karta hai. Super Admin ka step-by-step workflow:

Step 1: Store Provisioning	Super Admin navigates to Stores → Add Store . Inputs Store Name (<i>Sharma Medical Store</i>), City (<i>Varanasi</i>), Drug License, GSTIN. System auto-generates MED-000001.
Step 2: Store Owner Creation	Super Admin navigates to Store Owners → Add Owner . Enters Name (<i>Rahul Sharma</i>), Email (<i>rahul@gmail.com</i>), links to <i>Sharma Medical Store</i> , sets temporary password.
Step 3: Plan Selection	Super Admin verifies available plans under Subscription Plans (e.g. <i>Professional Plan @ ₹999/month</i>).
Step 4: Subscription Assignment	Super Admin goes to Store Subscriptions → Assign Plan . Selects <i>Sharma Medical Store</i> , assigns <i>Professional Plan</i> for 1 month / 1 year.
Step 5: Payment Recording	Super Admin records the payment of ₹999 under Payments → Record Payment via UPI with transaction ID. Status marked as <i>Paid</i> .
Step 6: Tenant Dashboard Access	Rahul logs in at /store/login. He accesses his private tenant portal to manage medicines, inventory, billing, purchases, and sales in complete isolation.

4. Solo Owner & Walk-in Customer Flexibility

Medical store businesses varying scale ke hote hain. Architecture me ye do flexibilities inbuilt hain:

Scenario	Architectural Handling	System Rule
Single Owner Pharmacy (Owner khud sab manage karta hai)	Owner login karke directly Billing POS, Stock Entry, Purchase Order aur Reports execute karega.	Staff account create karna OPTIONAL hai. No mandatory staffing dependency.
Multi-Staff Pharmacy (Cashier, Pharmacist alag hain)	Owner apne panel se staff members create karke specific module permissions assign karega.	Owner remains supervisor; staff accounts are scoped to owner's store.

Walk-in Retail Sales (OTC Customers)	POS counter par fast checkout ke liye customer mobile/name optional hai. Bill standard 'Walk-in Customer' record par ban sakta hai.	Customer ledger record tabhi compulsory hoga jab udhaar (credit) ya prescription history maintain karni ho.
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5. Super Admin Navigation Sitemap Hierarchy

Core Administrative Menu	Target URL Route	Primary Functional Objective
■ Dashboard	/super-admin/dashboard	Global SaaS metrics, revenue overview & recent activities
■ Stores	/super-admin/stores	Medical store provisioning, legal compliance & status control
■ Store Owners	/super-admin/store-owners	Owner account registration, store association & credentials
■ Subscription Plans	/super-admin/subscriptions/plans	Plan packaging, feature toggles & quota limit configuration
■ Store Subscriptions	/super-admin/subscriptions/stores	Assigning subscriptions, trial rules & renewal tracking
■ Payments	/super-admin/payments	Tracking transaction receipts, payment methods & statuses
■ Reports & Analytics	/super-admin/reports	Store growth velocity, revenue charts & CSV export tools
■ Notifications	/super-admin/notifications	Platform announcements, targeting & scheduling dispatch
■■ System Settings	/super-admin/settings	Global platform parameters, branding, security & maintenance
■ Audit Logs	/super-admin/audit-logs	Immutable security event ledger & before/after diff inspector

6. Golden Principles to Remember

Key Architecture Takeaways:

- Super Admin** = Poore platform ka global administrator (never belongs to a single store).
- Store Owner** = Apne medical store ka private owner (has full tenant isolation).
- Staff Management** = Optional capability (Solo owners can operate without staff).
- Customer Record** = Optional on counter POS (Walk-in sales do not require mandatory registration).
- Audit Logging** = Comprehensive, automatic, and read-only for compliance and governance.